

Titanic Dataset - Exploratory Data Analysis (EDA)

Introduction

This report presents an Exploratory Data Analysis (EDA) of the Titanic dataset. The objective is to extract insights using descriptive statistics and visualizations to understand survival patterns and key influencing factors.

Data Overview

The dataset contains information about Titanic passengers including demographics, ticket details, and survival outcome. The training set has 891 rows and 12 columns, while the test set has 418 rows and 11 columns.

Key Findings from EDA

- Females had a much higher survival rate compared to males.
- Passengers in higher classes (1st class) had significantly better survival chances.
- Younger passengers, especially children, had better survival outcomes.
- Higher fares were positively correlated with survival likelihood.
- Passengers embarked from port 'C' showed slightly better survival compared to 'S'.

Summary

The exploratory analysis highlights that survival was strongly influenced by gender, class, and fare. Women, children, and wealthier passengers were more likely to survive. These findings suggest socioeconomic and demographic factors played a critical role in survival.